

[Home](#) > [My courses](#) > [S1-25_MERGEDACIC](#) > [Quiz 1](#)**Status** Finished**Started** Sunday, 14 September 2025, 9:24 PM**Completed** Sunday, 14 September 2025, 9:34 PM**Duration** 9 mins 13 secs**Grade** 5.00 out of 5.00 (100%)**Question 1**

Correct

Mark 0.25 out of
0.25

Which of the following best describes the task environment in a game-playing AI agent designed to play chess,?

- ☐ a. A stochastic environment where randomness plays a significant role
- ☒ b. A fully observable and discrete environment with deterministic outcomes ✓
- ☐ c. A partially observable and continuous environment
- ☐ d. A real-time environment requiring constant updates and sensory input



The correct answer is: A fully observable and discrete environment with deterministic outcomes

Question 2

Correct

Mark 0.25 out of
0.25

In an AI-powered personal assistant that manages your daily schedule and recommends activities, how does the agent choose which activities to suggest next?

- ☐ a. By mapping directly from the user's requests to pre-programmed actions
- ☒ b. By maximizing the expected utility based on the user's preferences and priorities ✓
- ☐ c. By generating and following a rigid plan of action without considering user feedback
- ☐ d. By following fixed rules and ignoring user preferences

The correct answer is: By maximizing the expected utility based on the user's preferences and priorities

Question 3

Correct

Mark 0.25 out of 0.25



In the task environment, the classification into 'Deterministic' and 'Stochastic' primarily refers to which aspect of the agent's interaction with the environment?

- ☐ a. Sensor configuration
- ☒ b. Nature of actions and their consequences ✓
- ☐ c. State changes in the environment
- ☐ d. Agent's decision-making model

The correct answer is: Nature of actions and their consequences

Question 4

Correct

Mark 0.25 out of 0.25

In a food delivery drone, which of the following represents the Performance measure?

- ☒ a. The number of successful deliveries made on time ✓
- ☐ b. The sensors that detect obstacles during flight
- ☐ c. The battery level of the drone
- ☐ d. The accuracy of the drone's GPS system



The correct answer is: The number of successful deliveries made on time

Question 5

Correct

Mark 0.25 out of 0.25

A vacuum cleaner agent has the task environment where, at each time step, each clean square has 10% chance of becoming dirty. The task environment is

- ☒ a. Stochastic ✓
- ☐ b. Episodic
- ☐ c. Static
- ☐ d. Deterministic

The correct answer is: Stochastic

Question 6

Correct

Mark 0.25 out of 0.25

In the context of informed search algorithms, what does the heuristic function ($h(n)$) represent?

- ☒ a. The estimated cost from node n to the goal ✓
- ☐ b. The total cost from the start node to the goal node
- ☐ c. The number of nodes expanded so far
- ☐ d. The exact cost from the start node to node n

The correct answer is: The estimated cost from node n to the goal

Question 7

Correct

Mark 0.25 out of 0.25

In an electric vehicle (EV) management system, which of the following would be monitored by a sensor to influence decision-making?

- ☐ a. The engine's speed and power output
- ☐ b. The road conditions during the drive
- ☐ c. The navigation system's route suggestion
- ☒ d. The battery level of the vehicle ✓



The correct answer is: The battery level of the vehicle

Question 8

Correct

Mark 0.25 out of 0.25

In PEAS, if the agent is a virtual assistant designed to manage email correspondence, the Environment will be

- ☒ a. Email content and user requests ✓
- ☐ b. The user's satisfaction with the responses
- ☐ c. The algorithms used for email sorting
- ☐ d. The email client interface

The correct answer is: Email content and user requests

Question 9

Correct

Mark 0.25 out of 0.25

In the context of agent architectures, which component is responsible for executing the actions decided by the agent?

- ☒ a. Actuator system ✓
- ☐ b. Communication system
- ☐ c. Perceptual system
- ☐ d. Decision-making system

The correct answer is: Actuator system

Question 10

Correct

Mark 0.25 out of 0.25

In the context of a self-driving delivery robot, which aspect of the PEAS framework is responsible for determining the success of the robot's navigation and delivery tasks in complex urban environments?

- ☐ a. The environment's layout and traffic conditions
- ☒ b. The performance measure, based on timely delivery and avoidance of hazards ✓
- ☐ c. The actuators controlling the robot's movement through intersections
- ☐ d. The robot's ability to accurately detect obstacles and pedestrians



The correct answer is: The performance measure, based on timely delivery and avoidance of hazards

Question 11

Correct

Mark 0.25 out of 0.25

Which of the following scenarios correctly identifies the Environment in the PEAS framework for a chess-playing AI agent?

- ☐ a. The rules of the game
- ☐ b. The evaluation function that checks how good a move is
- ☐ c. The search algorithm used to determine the next move
- ☒ d. The opponent's moves and the current board state ✓

The correct answer is: The opponent's moves and the current board state

Question 12

Correct

Mark 0.25 out of 0.25

What is the primary difference between Greedy Best-First Search and A* Search?

- ☐ a. Greedy Best-First Search always finds the optimal solution, while A* Search does not.
- ☒ b. Greedy Best-First Search uses only the heuristic function, while A* Search uses both the heuristic and the cost function. ✓
- ☐ c. A* Search is faster than Greedy Best-First Search.
- ☐ d. A* Search is more memory-efficient than Greedy Best-First Search.

The correct answer is: Greedy Best-First Search uses only the heuristic function, while A* Search uses both the heuristic and the cost function.

Question 13

Correct

Mark 0.25 out of 0.25

In a hospital robot tasked with delivering medications, select the best task environment from the following.

- ☐ a. A fixed grid-like path with no obstacles
- ☒ b. A dynamic environment with unpredictable human movement and changing room layouts ✓
- ☐ c. A simulated environment with virtual patients
- ☐ d. A structured lab environment with pre-determined paths

The correct answer is: A dynamic environment with unpredictable human movement and changing room layouts

Question 14

Correct

Mark 0.25 out of 0.25

The peak in a Hill climbing algorithm is known as

- ☒ a. Local Maxima ✓
- ☐ b. Shoulder
- ☐ c. Ridge
- ☐ d. Plateau

The correct answer is: Local Maxima

Question 15

Correct

Mark 0.25 out of 0.25

In which scenario is Depth-First Search typically used instead of A* Search or other informed search strategies?

- ☐ a. When edge costs are important and need to be minimized
- ☐ b. When the heuristic function is very accurate
- ☒ c. When memory efficiency is a higher priority than finding the optimal path ✓
- ☐ d. When the goal is to find the shortest path in a large search space

The correct answer is: When memory efficiency is a higher priority than finding the optimal path

Question 16

Correct

Mark 0.25 out of 0.25

Straight Line Distance heuristic is

- ☐ a. Optimal and consistent
- ☒ b. Admissible and consistent ✓
- ☐ c. None of the mentioned
- ☐ d. Optimal and admissible



The correct answer is: Admissible and consistent

Question 17

Correct

Mark 0.25 out of 0.25

Competitive environment is most suitable for _____ search problems.

- ☒ a. Adversarial ✓
- ☐ b. Uninformed
- ☐ c. Informed
- ☐ d. Local

The correct answer is: Adversarial

Question 18

Correct

Mark 0.25 out of 0.25

In an AI agent that manages climate control in smart homes, what is the key characteristic of its task environment?

- ☒ a. A continuous environment where temperature and humidity levels change gradually ✓
- ☐ b. A static environment with unchanging temperature preferences
- ☐ c. A single-agent environment with fixed goals and no external influences
- ☐ d. A competitive environment where the agent competes against other agents

The correct answer is: A continuous environment where temperature and humidity levels change gradually

Question 19

Correct

Mark 0.25 out of 0.25

The functioning of a goal-based AI agent can be described as.....?

- ☐ a. It reacts solely based on predefined rules without considering future states
- ☒ b. It takes actions by evaluating the potential outcomes and determining the actions that achieve a specific goal ✓
- ☐ c. It acts based on probabilistic models, where actions are chosen randomly but within defined constraints
- ☐ d. It optimizes its behavior based on a utility function to maximize overall performance

The correct answer is: It takes actions by evaluating the potential outcomes and determining the actions that achieve a specific goal

Question 20

Correct

Mark 0.25 out of 0.25

For a stock trading AI agent, select the suitable the complex nature of its task environment, considering real-time uncertainty and incomplete market data?

- ☐ a. Deterministic with fixed outcomes
- ☐ b. Controlled with predetermined stock prices
- ☐ c. Fully observable with no hidden factors
- ☒ d. Partially observable and stochastic with market volatility ✓

The correct answer is: Partially observable and stochastic with market volatility

[← Previous](#)
[Past Question Papers](#)

[Next →](#)
[Assignment 2 - - Problem statement 1](#)

Contact for support : support@wilp.bits-pilani.ac.in

@2025 Copyright BITS Pilani WILP

